

Morphology-Conditioned Physics-Based Motion Imitation Across 128 Body Shapes

Abstract

Physics-based motion imitation must accommodate both the diversity of human movement and differences in body geometry and dynamics. We construct a text-associated, multi-morphology motion corpus from AMASS-derived HumanML3D clips using HUMOS, pairing shape-specific reference motions with matching SMPL-based simulation assets. The corpus contains 20,951 clip identities, including mirrored clips, across 128 body configurations, yielding 2,681,728 generated variants. Source captions remain associated metadata rather than controller inputs. A contact-aware kinematic refinement procedure reduces reference artifacts while retaining explicit limits on pose and root changes. We train one morphology-conditioned policy directly on the refined multi-shape references, using temporal slot/type attention and a streamed training pool. Deterministic, clip-grouped splits and morphology-matched evaluation distinguish held-out motion tracking from training-pool monitoring. Offline evaluation of the selected checkpoint reports 98.47% success on the 1,048-clip validation split and 99.14% on the independent 1,048-clip test split, each with one selected body configuration per clip and a capped horizon; these are trained-body, one-shape-per-clip results, not all-shape or unseen-body tests. An eight-run grid of four temporal architectures crossed with refined and unrefined references, on a 150-clip development set, selects temporal slot/type attention without adaLN-Zero. Residual failures on both splits are analysed against inherited captions and concentrate on ground-contact transitions and whole-body acrobatics. A separate analysis asks whether body shape and motion type jointly set physical control demand, replaying the trained policy across all 19,200 (clip, shape) pairs of the 150-clip development set: after correcting a normalization artifact that would otherwise have produced a spurious positive result, we find no statistically detectable evidence that highly dynamic motions are more shape-sensitive than quasi-static ones for this checkpoint. A further test on 32 held-out body shapes finds 28 of 32 track at trained-body levels with no decline across distance from training, while the remaining 4 fail severely in a gender-flip pattern consistent with an asset artifact rather than a genuine limit, reported as an open anomaly rather than averaged away.

1 Introduction

A motion-imitation controller should reproduce an action without assuming that every character has the same body. Changes in limb lengths, mass distribution, and collision geometry alter both the reference trajectory and the dynamics required to follow it: replaying an identical joint sequence across bodies does not preserve foot placement, clearance, or balance, while optimizing only spatial agreement can obscure whether the intended articulation survives. Learning across body shapes therefore requires a consistent relationship between source motion, shape-specific reference, and simulated body.

Existing work provides large motion-capture collections, body-shape-conditioned motion generation, and scalable physics-based control [1, 2, 3, 4, 5, 6]. We build on these components to study a practical question: how can one policy imitate a large collection of shape-specific motions while

preserving traceable data identities and a meaningful evaluation protocol? Our contribution is this integration and its empirical assessment, not a claim that morphology conditioning or attention is new.

Our data pipeline extends AMASS-derived HumanML3D clips to 128 body configurations. HUMOS generates the corresponding reference motions; SMPLSim constructs simulation assets from the same body parameters. A conservative refinement pass smooths rotations, reduces stance sliding through root translation, and enforces clearance of the modeled collision geometry. The inherited descriptions connect each source action to its generated variants, supporting paired analysis and future language-based uses, although the controller studied here consumes reference motion, not text.

The selected controller combines current state, dilated state history, and future reference targets through temporal slot/type attention, with body parameters and a previous action concatenated into separate actor and critic heads. The main experiment trains directly on multiple shapes; neutral-body pretraining and mixed-source reward training were explored during development but are not part of the final method. A 150-clip development subset makes design exploration affordable and precedes full-scale training on a versioned split.

This paper makes four contributions:

1. A derived, text-associated multi-morphology corpus, with an explicit construction procedure linking source clips, generated references, body assets, and reproducible splits. Its captions are inherited rather than newly annotated for each body.
2. A single morphology-conditioned imitation system operating across 18,855 training clips and 128 body configurations, integrating temporal observations, matching simulation assets, reference refinement, and memory-bounded motion streaming.
3. An empirical protocol separating architecture, reference quality, and evaluation correctness: an eight-run architecture/reference grid on a 150-clip development set, a full-scale validation and independent test evaluation of the selected checkpoint, a caption-grounded analysis of the residual failures, and a generalization test on 32 body shapes excluded from training entirely.
4. A physical analysis of how body shape and motion type jointly set control demand, replaying the trained policy across every clip/shape pair of the development corpus to test whether dynamic motions are more shape-sensitive than quasi-static ones, rather than simply assuming it.

The selected checkpoint reaches 98.47% success on validation and 99.14% on the untouched test split under the protocol in Section 5.3. These results concern held-out clip identities on trained body configurations, with one body evaluated per clip. Section 6.5.1 extends the claim to bodies excluded from training entirely: 32 new shapes drawn from the trained coefficient range, of which 28 track successfully at the trained-body level with no decline across the evaluated distance from training, while 4 fail severely in a pattern pointing to an asset artifact rather than a genuine shape effect. Section 7 tests the morphology $\times$ motion-class interaction directly and finds no statistically detectable effect for this checkpoint, once a normalization artifact in the naive comparison is corrected.

2 Related Work

2.1 Physics-based motion imitation

DeepMimic demonstrated reference-guided reinforcement learning for simulated character skills [4]. Perpetual Humanoid Control developed robust humanoid imitation for real-time simulated avatars [5]. These approaches motivate learning a feedback controller rather than directly replaying a kinematic trajectory. A parallel line adds adversarial style priors and reusable low-level skills on top of this feedback-control paradigm: AMP conditions style on an adversarial motion prior [20], ASE learns large-scale reusable adversarial skill embeddings [21], CALM adds directable conditional latent control [22], and MaskedMimic unifies control modalities through masked motion inpainting [23]. Subsequent single-body trackers scaled this feedback-control approach in coverage and deployability: PULSE distills a low-dimensional, high-coverage motion representation for general humanoid control [15]; OmniH2O and ExBody2 target dexterous, deployable whole-body teleoperation and expressive control on physical humanoids [16, 17]; GMT uses a motion mixture-of-experts to track a broad motion distribution with one policy [18]. Each of these typically targets a fixed robot platform or morphology while scaling the motion distribution; we instead fix the tracking task and scale the body, holding 128 configurations under one policy. We implement our experiments in ProtoMotions [6] with Isaac Gym [7], using PPO optimization [8]. Our contribution is the multi-morphology data and control workflow and its evaluation, rather than a replacement for these simulation or learning frameworks.

2.2 Morphology-conditioned control and retargeting

Body-shape variation has already been studied in physics-based control: Won and Lee learned a parametric controller that accommodates variation in character height, weight, and proportions [10]. A separate line of work learns policies that transfer across changes in agent *topology* – limb count and skeletal structure – through shared modular sub-policies [19], a different axis from the continuous anthropometric variation studied here. More recent work continues both threads: AdaptNet adapts a physics-based controller across body-shape changes via policy adaptation [24], while a line of morphology-conditioned reinforcement learning generalizes one policy across many (typically non-humanoid) robot morphologies using contextual modulation [25], hypernetwork distillation [26], and shared modular recurrence [27]. Morphology conditioning is therefore an established idea; our contribution is not the concept but a system that examines one controller over a large set of paired motions and bodies, with shape identity maintained throughout data generation, simulation, and evaluation.

Table 1 summarizes how this positions against the work discussed above. No shared benchmark or task spans these systems, so the comparison is qualitative rather than a head-to-head evaluation; to the best of our knowledge, none of them trains and evaluates one policy across a paired, multi-shape motion corpus at anywhere near this scale.

Kinematic retargeting and dynamic tracking address different requirements. SMPL provides a common parameterized body representation [11], and SMPLSim constructs simulation-ready assets from these models [12]. A shared skeleton makes poses comparable, but does not make global positions or required actuation independent of shape. Matching a reference to its intended simulation asset is consequently part of the task definition, not an optional evaluation convenience. Data-driven kinematic controllers such as Mode-Adaptive Neural Networks [28] and DeepPhase [29] achieve high-quality motion blending without physics simulation or shape conditioning, underscoring that kinematic motion quality and physically/shape-consistent actuation are separate problems.

Table 1: Qualitative positioning against the physics-based imitation and morphology-conditioned control work discussed above. “Body configs” counts distinct trained body shapes evaluated under one policy, not motion coverage; entries reflect how each work is characterized in this section, not a full independent audit of those papers.

Method	Task focus	Morphology axis	Body configs
DeepMimic [4]	Reference-guided skill imitation	None (single body)	1
PHC [5]	Real-time humanoid imitation	None (single body)	1
PULSE [15]	Universal low-dimensional motion representation	None (single body)	1
OmniH2O [16]	Whole-body teleoperation	None (single physical robot)	1
ExBody2 [17]	Expressive whole-body control	None (single physical robot)	1
GMT [18]	Broad motion tracking (mixture-of-experts)	None (single body)	1
Won & Lee [10]	Parametric body-shape control	Continuous (height, weight, proportions)	Parametric range; count not specified
Shared modular policies [19]	Cross-topology transfer	Topology (limb count/structure), not anthropometric shape	N/A – topology axis
AdaptNet [24]	Physics-based policy adaptation	Body-shape changes via adaptation, not joint multi-shape training	N/A – adaptation, not shared training
ModuMorph [25]	Universal morphology control	Robot topology/limb configuration (non-humanoid)	Hundreds of UNIMAL morphologies
This work	Multi-morphology motion imitation	Continuous anthropometric shape (SMPL β), matched simulation assets	128 , over 20,951 paired clips

2.3 Text-motion data, shape-conditioned generation, and refinement

AMASS consolidates motion capture in a common body-model representation [1]. HumanML3D associates processed motion segments with natural-language descriptions [2]. Other text-motion resources take different design points: the KIT Motion-Language Dataset pairs shorter clips with structured annotations [30], BABEL instead attaches discrete per-segment action labels [31], and Motion-X scales to expressive whole-body capture with a larger and more diverse corpus [32]. Generative text-to-motion models built on such resources range from VAE-based approaches [33] and tokenized sequence-to-sequence generation [34] to diffusion models [35, 36]; HUMOS instead conditions generation on body shape rather than text alone. We inherit HumanML3D’s source identities and descriptions rather than claiming new motion capture or shape-specific language annotation.

HUMOS generates identity-conditioned human motion using a cycle-consistent learning formulation [3]. We use its pretrained model to obtain shape-specific references rather than as the physics

Motion	AMASS $\rightarrow$ HumanML3D preprocessing and clip extraction $\rightarrow$ HUMOS conditioned on (g, β) $\rightarrow$ MotionLib conversion and reference refinement $\rightarrow$ per-clip motion variants.
Body	SMPL with the same (g, β) $\rightarrow$ SMPLSim $\rightarrow$ matching skeleton, collision geometry, mass/inertia, and MJCF asset identity.
Text	HumanML3D source descriptions $\rightarrow$ clip-ID lookup associated with the generated variants (metadata only; not a policy input).
The body assets are used for motion conversion, shape-specific forward kinematics, ground-clearance correction, and simulation. Refined reference motion and matching simulated state feed one shared policy.	

Figure 1: Dataset and asset construction. The body parameters are shared across the motion and body paths; text follows a separate identity-association path.

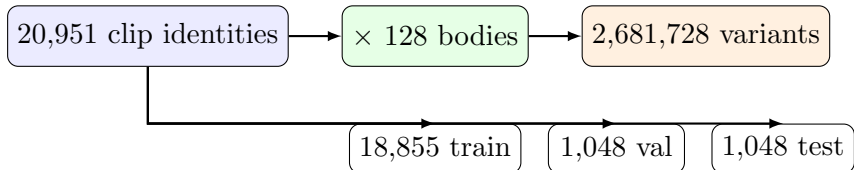


Figure 2: Scale and split structure of the derived corpus. Splitting is by clip identity, so all 128 body variants of a clip remain in the same partition.

controller or a text-conditioned policy. The resulting sequences still require conversion to the simulator’s skeleton and coordinate conventions, and their dynamic feasibility must be assessed in that simulator; our refinement is a deliberately limited kinematic post-process, not a new generative model or a full trajectory optimizer.

The controller uses standard Transformer operations [13], with learned slot and type embeddings identifying temporal positions and observation sources. An ablation also investigates actor-only morphology modulation with adaLN-Zero, adapting the zero-initialized conditioning design used in Diffusion Transformers [14] as a conditioning mechanism rather than for diffusion-based action generation.

3 Multi-Morphology Text-Associated Motion Dataset

We construct a derived corpus in which a source clip identity connects a motion description, 128 generated motion variants, and their matching simulation assets. A *clip* denotes a source segment, including a mirrored segment when present; a *variant* denotes that clip on one body configuration, and we keep the two units distinct throughout when reporting dataset size or evaluation coverage. Figure 1 shows the construction paths and Figure 2 the resulting scale and split structure.

3.1 Source motion and text provenance

We use the AMASS-derived motion portion of HumanML3D [1, 2], with the 18 source collections listed in Appendix A.1 – a subset of AMASS, not its entirety. Source SMPL+H sequences are processed at 20 Hz, including the Z-up to Y-up coordinate conversion used by the HumanML3D/HUMOS preprocessing pipeline. Clip extraction, temporal cropping, and mirroring preserve the HumanML3D identities, and the preparation also includes treadmill/skating cleanup.

The generation pipeline produced 22,459 candidate clip identities, counting original and mirrored clips. A subsequent text-based filter excluded 1,508 clips identified as requiring external objects or environmental support, leaving 20,951 identities for the flat-ground imitation task. This filter is a task-scope heuristic rather than a feasibility proof: some retained references may still involve implicit support the filter did not catch. The retained and excluded ID lists provide the audit trail for that decision.

Source descriptions are available through a separate clip-ID lookup and the HumanML3D annotation metadata used by HUMOS, associated with the variants of a source clip rather than newly annotated for each generated body. A coarse coverage check against a single-caption-per-clip lookup finds a caption for all 20,951 clip identities, though semantic fidelity is not separately established. Caption inheritance is imperfect for mirrored clips in particular: of the 9,579 mirrored identities, 22% carry a caption byte-identical to their un-mirrored source, and 40% contain an explicit left/right or clockwise term that mirroring inverts, so a mirrored clip’s caption cannot be corrected by simply stripping a mirror prefix. Segment timing and multi-annotation coverage against the full HumanML3D annotation set are left for future auditing.

3.2 Body sampling and shape-specific generation

Let a body configuration be $b = (g, \beta)$, where g identifies the female or male SMPL body-model variant and $\beta \in \mathbb{R}^{10}$ contains shape coefficients. We sample

$$\beta_{ij} \sim \mathcal{U}(-3, 3), \quad i = 1, \dots, 64, \quad j = 1, \dots, 10, \quad (1)$$

using NumPy’s `default_rng(46)`, and cross these 64 vectors with the two body-model variants, giving 128 body configurations rather than 128 independently sampled beta vectors. The actual coefficient arrays and stable beta keys identify the bodies. Uniform sampling of model coefficients is a controlled source of variation, not a claim of representing the distribution of human populations.

For each source clip, we provide the processed pose/root motion features and the target body parameters to the pretrained HUMOS model [3], using its root-relative 6D root and joint rotation configuration with shape and body-model conditioning, and retain the generated pose and root sequences for conversion. HUMOS is an existing identity-conditioned model; neither its training nor a text-to-motion generator is a contribution of this work. The checkpoint used in the documented pipeline is identified in Appendix A.4.

In parallel, SMPL bodies [11] with identical parameters are converted to MJCF assets using SM-PLSim [12]. The asset-generation configuration uses mass estimation and the configured hand/foot simplifications (`real_weight`, `replace_feet`, `remove_toe`, and `freeze_hand`). The resulting imitation representation contains 24 rigid bodies and 69 actuated joint coordinates, each asset carrying its own collision proxies, joint ranges, and estimated masses/inertias – simulation-model estimates rather than measurements of corresponding human participants.

3.3 Motion conversion and reference refinement

The generated sequences are converted to ProtoMotions’ Z-up simulation convention and resampled from 20 to 30 Hz. Forward kinematics (FK) uses the matching asset rather than a single neutral skeleton. The packaged representation includes global body positions and rotations, local rotations, joint positions, linear/angular and joint velocities, and contact labels. The earlier corpus already included frame-zero grounding; the newer refinement addresses artifacts throughout each sequence.

Refinement is applied independently to each variant in the following order. First, stored ankle/toe contacts are validated, or contacts are detected from foot position and velocity if usable

Table 2: Current corpus and split sizes. Variants are generated combinations, not independent captures. Every clip has 128 body configurations.

Partition	Clip identities	Motion variants
Training	18,855	2,413,440
Validation	1,048	134,144
Test	1,048	134,144
Total	20,951	2,681,728

labels are unavailable. Short gaps are closed and very short contact runs are removed. These labels define candidate stance intervals; they are not measured ground-reaction forces.

Second, joint rotations are temporally filtered using sign-continuous quaternion representations, normalization, and a bounded geodesic correction. Contact-transition frames divide the sequence into segments so filtering does not average across a change of stance. A transition taper and a limited smoothing blend further restrict changes near those boundaries. Post-smoothing FK then recomputes the positions used by subsequent corrections.

Third, a conservative root-translation correction reduces horizontal stance sliding. For each contact interval, an anchor is formed from the foot’s median horizontal position. A least-squares correction seeks a common root XY shift that brings the active feet closer to their anchors; the shift is temporally regularized and capped. Multiple contacts can disagree, so this root-only operation does not guarantee exact foot locking and does not solve for new joint angles.

Fourth, we evaluate the lowest point of the matching collision geometry at every frame. With floor height zero and margin ϵ_z , the required lift is

$$\ell_t^{\text{req}} = \max\left(0, \epsilon_z - \min_{x \in \mathcal{C}_b(t)} x_z\right), \quad (2)$$

where $\mathcal{C}_b(t)$ denotes the modeled collision geometry for body b . A smoothed lift envelope is constrained to remain above this requirement. This prevents penetration of the checked geometry in the reference sequence; it does not establish balanced support, realistic impacts, or freedom from self-collision.

Finally, FK and velocities are recomputed from the corrected root and rotations, independently within each motion. This avoids inconsistent position/velocity tensors and artificial derivatives across clip boundaries. The default thresholds, filter windows, and correction bounds are given in Table 14.

The refinement objective is modest: reduce visible and measurable kinematic artifacts while preserving the generated motion. We therefore assess it with both artifact measures (sliding, penetration, and jerk) and fidelity measures (rotation/root changes and retained cross-shape variation), since improved kinematics alone would not demonstrate improved policy learning or dynamic feasibility (Section 6.4).

3.4 Packaging, scale, and deterministic splits

Each per-clip file contains the 128 variants, their frame counts and sampling rates, and the clip/asset identities needed to recover the corresponding body parameters. Captions remain a separately associated metadata resource rather than a field of every training tensor file. This packaging allows the training system to stream clip files without downloading the complete corpus at startup.

We use the versioned `hhi_stage2_v1` split with seed 42 and approximate 90/5/5 proportions (Table 2). All variants of a clip stay together, and original/mirrored partners are assigned as a group. The 150 development clips are reserved for training, together with their mirror partners. Training accepts explicit training and validation manifests; test-manifest membership is not loaded by the training workflow. Split hashes and version information accompany the run and checkpoints.

These splits test held-out clip identities on the trained body set, but they are not demonstrated to be disjoint by subject or original capture recording, since different segments can still share a source recording, which limits claims about broader motion generalization.

3.5 Current resource scope

The current resource combines generated references, matching assets, inherited text associations, and reproducible splits, but is not a fully audited text-and-motion benchmark release. Caption completeness and alignment, body-specific language, licensing and redistribution conditions, and shape-conditioned semantic fidelity are left for future work, and can be developed independently of the reference-conditioned controller evaluated here.

4 Morphology-Conditioned Motion Imitation

4.1 Task, simulation, and actuation

For a clip c and body configuration b , let $\hat{s}_{c,b}(t)$ be the refined reference state and s_t the simulated state of the matching asset. We learn a shared policy $\pi_\theta(a_t | o_t, b)$ that tracks this reference on flat ground. The body assigned to an environment is fixed, while its reference motion can change on reset. No language input or external scene object is supplied.

Experiments use the `smp1_mor` assets in Isaac Gym through ProtoMotions [7, 6]. Simulation runs at 60 Hz with two substeps and control decimation two, giving a 30 Hz policy rate. The policy outputs a raw action $a_t \in \mathbb{R}^D$, where $D = 69$. Joint targets are

$$q_t^{\text{target}} = q^{\text{offset}} + q^{\text{scale}} \odot \tanh(a_t), \quad (3)$$

with offsets and scales derived from the configured joint ranges. These targets drive the simulator’s built-in position PD controller directly, rather than as a residual added to the reference pose.

PD stiffness, damping, and effort limits are scaled per joint by the mass of its associated rigid body relative to that body in the reference asset – a body-level mass scaling rather than a single scale based on total character mass, which does not by itself guarantee dynamically equivalent actuation across geometrically different bodies. Base gains and limits are listed in Table 15.

4.2 Observation design

The observation combines present state, history, future reference targets, a historical action, and explicit morphology information. Let $B = 24$ denote the number of simulated bodies. The state representation contains root height above ground, root-relative body positions excluding the root, 6D body rotations, and linear/angular velocities. Positions and velocities use a heading-aligned frame, and rotations remove the root heading. This removes dependence on arbitrary global heading and horizontal origin while retaining height and body configuration.

We include seven historical states at control-step offsets $\{1, 2, 3, 4, 8, 16, 32\}$, covering approximately 1.07 s, and four future reference slots at $\{1, 2, 4, 8\}$, covering approximately 0.27 s. Each historical state uses its own root/heading frame. Future slots contain target positions relative to

Table 3: Observation inputs supplied to both actor and critic. Dimensions are per slot. Positions/heights use meters, linear velocities m/s, angular velocities rad/s, and 6D rotations are dimensionless. Raw actions and morphology coefficients are not joint-angle measurements.

Input	Dimension	Slots	Representation and path
Current state	$15B - 2 = 358$	1	Height; $3(B - 1)$ positions; $6B$ rotations; $3B$ linear and $3B$ angular velocities. Current-state encoder.
History	358	7	Same state representation at the seven past offsets. Shared history encoder.
Future reference	$24B = 576$	4	Target and relative poses ($18B$), plus velocity differences ($6B$). Shared future encoder.
Historical action	$D = 69$	1	A raw previous policy-action vector, before tanh and PD scaling. Bypasses attention.
Morphology	11	1	$[g, \beta_1, \dots, \beta_{10}]$ from the assigned asset. Bypasses attention.

the current root, heading-aligned target rotations, position differences from the current body state, relative rotations $R_{t,j}^{-1} \hat{R}_{t+k,j}$, and heading-aligned linear/angular velocity differences. Thus relative rotations are expressed with respect to the current body rotation, not as another heading-frame position feature. Table 3 lists the resulting input dimensions and slot counts.

The temporal context is designed to expose both recent changes and more distant motion information, with the future targets providing a short anticipation window; we have not separately ablated these design choices. History is initialized consistently with reference-state resets using available historical reference states, falling back to single-state initialization when a trajectory history is unavailable. Contact flags, an explicit phase scalar, text, and estimated physical feature vectors are not part of the selected observation.

4.3 Temporal slot/type attention

We use three observation encoders: one for the current state, one shared across history slots, and one shared across future slots. Each is an MLP with two 256-unit ReLU hidden layers and a 256-dimensional token output. Inputs use running mean/standard-deviation normalization and a magnitude clamp of five.

The resulting 12 tokens comprise one current, seven historical, and four future tokens. For slot i , the Transformer input is

$$z_i = E_{\tau(i)}(x_i) + e_i^{\text{slot}} + e_{\tau(i)}^{\text{type}}, \quad i = 0, \dots, 11, \quad (4)$$

where $\tau(i)$ identifies the current/history/future source. Slot embeddings distinguish the configured temporal offsets, while type embeddings distinguish the observation sources. The Transformer has two layers, four attention heads, token width 256, and feed-forward width 1,024 [13]. Its current-state token supplies the pooled output after the configured output projection and activation.

The final head input is the concatenation of the 256-dimensional attention output, the historical raw action, and the 11-dimensional morphology vector. This 336-dimensional input also uses running normalization and clamp five. In the selected model, betas enter only through this concatenation: they are not separately rescaled, embedded as their own attention token, or used for adaLN modulation.

Actor and critic have separate token encoders, attention modules, slot/type embeddings, and MLP heads. The actor head has six hidden layers of width 2,896; the critic has four of width 1,024, all with ReLU activations. The outputs are a Gaussian action mean and a scalar value, respectively.

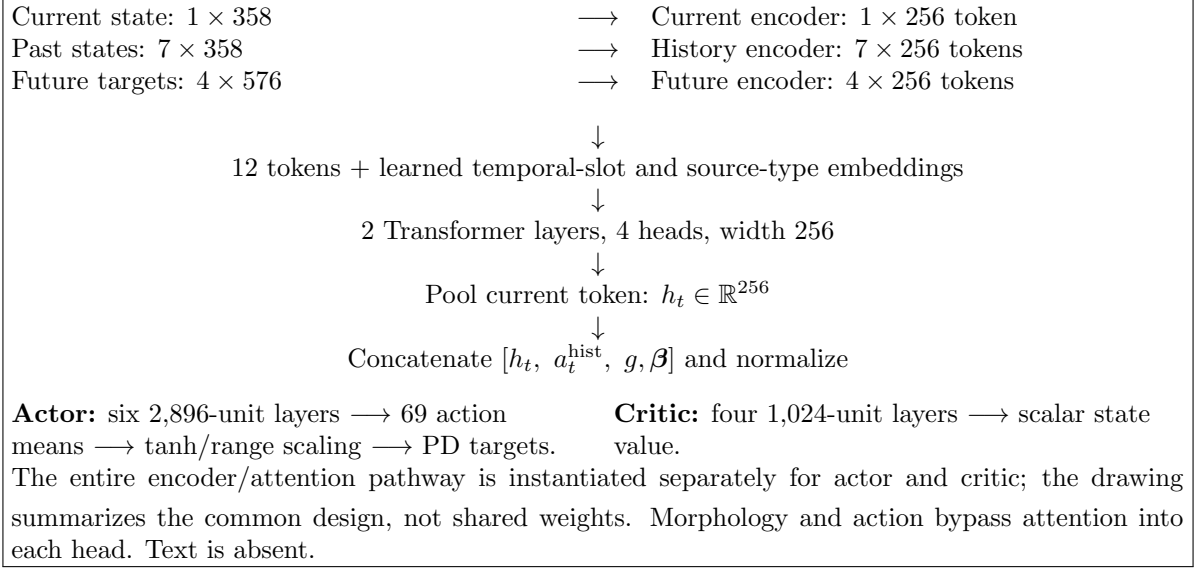


Figure 3: Observation-to-action data flow of the selected slot/type policy. Both networks receive the same input categories but have separate parameters.

PPO samples actions with fixed diagonal log standard deviation -2.9 ; evaluation uses the mean action. Figure 3 summarizes the flow.

4.4 Tracking reward, resets, and PPO

The training reward combines global body tracking and root-height tracking. For corresponding simulated and reference body positions $p_j, \hat{p}_j$, rotations $R_j, \hat{R}_j$, and velocities $v_j, \hat{v}_j$ and $\omega_j, \hat{\omega}_j$, define

$$e_p = \frac{1}{3B} \sum_{j=1}^B \|p_j - \hat{p}_j\|_2^2, \quad e_R = \frac{1}{B} \sum_{j=1}^B d_{\text{SO}(3)}(R_j, \hat{R}_j)^2, \quad (5)$$

$$e_v = \frac{1}{3B} \sum_{j=1}^B \|v_j - \hat{v}_j\|_2^2, \quad e_\omega = \frac{1}{3B} \sum_{j=1}^B \|\omega_j - \hat{\omega}_j\|_2^2, \quad (6)$$

$$e_h = (h - \hat{h})^2, \quad (7)$$

where $d_{\text{SO}(3)}$ is the geodesic rotation angle in radians. The coordinate-wise means in the position and velocity errors match the implemented reductions. The reward is

$$r_t = 0.5 \exp(-25e_p) + 0.3 \exp(-5e_R) + 0.1 \exp(-0.5e_v) + 0.2 \exp(-0.1e_\omega) + 0.2 \exp(-100e_h). \quad (8)$$

The weights are unnormalized (they do not sum to one), and this experiment uses no explicit effort, action-smoothness, or contact-matching reward term, and no source-dependent switch between canonical and HUMOS reward functions.

References are sampled only from variants matching the environment’s asset. Reference-state initialization starts at frame zero with probability 0.2; otherwise it uses a random valid reference

time. Training episodes have a maximum of 1,000 control steps and use the configured fall termination with a 0.15 m height setting and permitted foot-contact bodies. Tracking-error termination is disabled. This training rule is distinct from the evaluation failure threshold in Section 5.4.

We optimize using PPO [8], with 32-step rollouts, discount 0.99, generalized advantage estimation [9] parameter 0.95, policy-ratio clipping 0.2, and normalized advantages. Separate Adam optimizers use learning rates 2×10^{-5} for the actor and 10^{-4} for the critic. Other configuration values and actual run budgets appear in Table 4. Evaluation rollouts use deterministic inference without gradient tracking.

4.5 Direct multi-shape learning and streamed sampling

The final approach is direct multi-shape training on refined HUMOS references. A small static corpus supports initial development; the full-scale experiment streams per-clip files through `GlobalClipPool`. The training manifest is deterministically partitioned across GPU ranks. Each rank maintains a bounded resident pool containing all 128 variants of its selected clips. Environment-level motion sampling respects the assigned morphology.

At each scheduled rebuild, most resident slots are selected by explicit top priority, combining estimated clip difficulty and an exploration bonus. A fraction 0.2 is reserved for uniform random selection, maintaining rehearsal outside the highest-priority set. Difficulty estimates are updated from training-pool evaluations with an exponential moving average of rate 0.1 and a floor of 0.05, with exploration coefficient 1.0 – a training curriculum rather than uniform sampling of the complete corpus.

The resident pool is rebuilt every 256 epochs. The full-scale run initially used 256 clips per rank and was resumed with 128 after memory failures, with the local cache multiplier reduced to 1.0; these changes affect residency and memory demand only, not the underlying training split. Validation membership is fixed independently of pool rebuilding, and validation evaluation does not update training sampling weights, so we report resident-pool and validation metrics separately as they measure different populations.

5 Experimental Setup

5.1 Development subset and full-scale protocol

The development corpus contains 150 clips and all 128 body configurations, for 19,200 variants. These clips are reserved within the full training split and are not a held-out generalization benchmark. Small-scale experiments investigate architecture and reference-processing choices with a fixed simulation budget. The full-scale run uses the 18,855 training clips in Table 2, with validation membership fixed throughout.

The selected full-scale experiment is slot/type attention trained directly on refined references. Its code retains a historical `stage2` name, but this run does not use neutral pretraining followed by shape transfer. Table 4 records the shared optimization settings and the actual environment/batch sizes used after the initial memory-related launch adjustments. Run-specific resolved configurations take precedence over older launch examples.

5.2 Architecture and refinement controls

The corrected comparison is a four-architecture by two-reference-version grid, Table 5. The four architectures are a wide flat temporal MLP, basic temporal attention, attention with slot/type

Table 4: Training configuration and budgets. Environment and minibatch counts are per GPU. The full-scale frame count is a recorded stopping snapshot, not a budget matched to the development runs.

Setting	Corrected 150-clip runs	Full-scale selected run
Parallel GPUs per policy	1	6
Environments per GPU	4,096	2,048
PPO minibatch size per GPU	16,384	8,192
Rollout length	32 control steps	32 control steps
Mini-epochs per update	1	1
Actor / critic learning rates	$2 \times 10^{-5} / 10^{-4}$	$2 \times 10^{-5} / 10^{-4}$
Discount / GAE / ratio clip	0.99 / 0.95 / 0.2	0.99 / 0.95 / 0.2
Gradient norm clip	50	50
Evaluation cadence	200 epochs	256 epochs
Training seed	0 (all eight runs)	Recorded in saved run configuration
Reference residency	All 150 clips	256, later 128 clips per GPU
Training frames	786,432,000 per run (6,000 epochs)	Approximately 11.07×10^9 recorded
Status of reported evidence	Completed, Table 9	Epoch-28,170 checkpoint, Table 8

embeddings, and that last model plus actor-only adaLN-Zero. Each is trained once on refined references (A–D) and once on unrefined references (E–H); slot/type on unrefined references is labelled E rather than a separate letter for historical reasons. All eight runs use one seed, a single A40, the same 6,000-epoch budget, evaluation cadence, and shape-sampling seed.

All eight runs use seed 0 and one A40 GPU per policy. Morphology concatenation into both heads is retained in D and H; adaLN-Zero is additional actor conditioning, not a replacement for the critic or a two-stage schedule.

The models do not have equal parameter counts (Table 6). The grid is single-seed; the architecture ordering in Table 9 rests on the jerk difference, which recurs under both reference versions, and not on the success-rate column, which is treated as non-discriminating. The refined and unrefined pairs (A/F, B/G, C/E, D/H) are each evaluated against their own reference version, so they measure whether refined training costs policy quality, not whether it causally improves accuracy; the refinement argument is completed by the offline reference-artifact audit in Section 6.4 rather than by a common-reference controller re-evaluation. A causal comparison would require scoring refined- and unrefined-trained checkpoints on one shared reference version.

Earlier small-scale runs influenced the choice to begin full-scale slot/type training. The corrected reruns were launched later to reassess that choice; we do not present them as completed model selection preceding the full run.

5.3 Morphology-matched evaluation and checkpoint scope

Each reference is rolled out only in an environment using its exact asset identity (Figure 4). A selected reference therefore cannot be assigned to a different body simply because an environment slot is available. Multiple rollout batches may be needed when several references require the same body.

Routine monitoring selects one variant per clip. With seed $s = 42$, stable clip identity c , panel index p , and N_c available variants, the choice is

$$k(c, p) = (\text{uint64}_{\text{big}}(\text{SHA256}(s:c)_{0:8}) + p) \bmod N_c. \quad (9)$$

Here $s:c$ is the UTF-8 string formed as `seed:clip.id`. For scheduled monitoring, p is the integer

Table 5: Corrected 150-clip grid: four architectures $\times$ two reference versions, one seed each. Results are in Table 9.

Case	References	Architecture	Comparison purpose
A	Refined	Wide temporal MLP; beta concatenation	Temporal MLP baseline.
B	Refined	Basic temporal attention	Attention versus flattened temporal input.
C	Refined	Attention with slot/type embeddings	Explicit temporal/source identity.
D	Refined	C + actor-only adaLN-Zero	Morphology modulation with critic unchanged.
E	Unrefined	As C	Reference version at fixed architecture (slot/type).
F	Unrefined	As A	Reference version at fixed architecture (temporal MLP).
G	Unrefined	As B	Reference version at fixed architecture (attention).
H	Unrefined	As D	Reference version at fixed architecture (slot/type + adaLN).

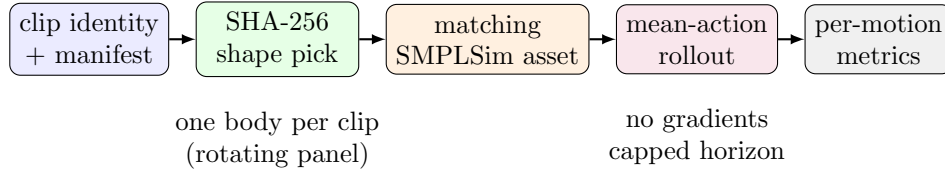


Figure 4: Standalone evaluation protocol. The clip list is fixed, while a reproducible hash-and-panel rule selects one body variant per clip; the selected reference and simulation asset therefore always match.

quotient of the checkpointed epoch and evaluation cadence. This produces deterministic staggered shape choices that rotate across evaluations and remain reproducible after resume. The validation *clip list* is fixed; the selected *shape panel* changes. Thus successive scores are not repeated measurements on an identical set of clip/shape pairs.

Evaluation uses the mean action, reference initialization, and no gradient tracking. It disables training termination and records errors over the valid reference horizon, capped at 600 control steps (20s at 30 Hz). Success therefore refers to the evaluated horizon, not necessarily the entire duration of a longer source clip.

Final model comparisons should freeze the checkpoint, clip/shape panel, reference version, simulator configuration, and seed. Validation is available for model selection and has been repeatedly monitored; it is not an untouched test set. The independent test manifest remains outside training and model selection; its one-time evaluation is reported in Section 6.2. The downloaded epoch-28,170 `last.ckpt` is distinct from the epoch-28,159 monitoring snapshot in Table 7; the standalone validation and test results therefore refer explicitly to the downloaded checkpoint.

5.4 Metrics and aggregation

For each evaluated clip/shape pair i , define its per-frame mean body position error

$$d_{i,t} = \frac{1}{B} \sum_{j=1}^B \|p_{i,t,j} - \hat{p}_{i,t,j}\|_2. \quad (10)$$

The configured success indicator is

$$S_i = \mathbf{1} \left[\max_{t < T_i} d_{i,t} \leq 0.5 \text{ m} \right], \quad (11)$$

where T_i is the evaluated horizon. This threshold is deliberately distinct from the smoother training reward. Passing it does not imply fine-grained pose accuracy or biomechanical realism.

We also report time-averaged mean body-position error, mean global body-rotation geodesic error, and the time average of the largest body-position error within each frame – the logged *mean maximum-joint error*, a per-frame worst-body statistic averaged over time rather than a single dataset-wide maximum. Global body-rotation error is a whole-body geodesic measure, not a local joint-angle/DOF error.

Within an evaluator rank, errors are averaged over valid frames per motion and then over motions, including failed rollouts. The distributed training logger aggregates scalar summaries across ranks; the monitoring table retains those logged values rather than reconstructing exact pooled counts. Final offline comparisons should aggregate saved per-motion records over unique clip/shape pairs, avoiding unequal weighting of partially filled batches or duplicate padded rollouts.

Trajectory smoothness uses the implementation’s windowed normalized-jerk score, with the definition and numerical regularizers in Appendix A.3. Lower scores indicate lower jerk under that fixed convention. The accompanying high-jerk statistic is the percentage of windows in which any body exceeds threshold 6,500; the logger calls it a frame percentage, but it is window-based.

Action increments require care. The current evaluator stores raw policy actions, before Equation (3). Their differences are policy-output changes, not physical joint-angle changes, even where logger field names label them radians or degrees. We omit those mislabeled angular values from the physical interpretation and from Table 7. PD-target changes, applied torques, and measured joint motion must be computed separately.

5.5 Generalization axes and compute accounting

Held-out clips on trained bodies, seen clips on unseen bodies, and held-out clips on unseen bodies are different evaluations. The main validation and test tables address only the first. A single rotating panel does not measure all 128 variants of every clip, and future cross-shape comparisons must report the actual body identities and number of evaluated pairs. The unseen-morphology evaluation in Section 6.5.1 covers the other two axes directly, on shapes drawn from the trained coefficient range but excluded from the 128 training configurations, with results stratified by distance to the nearest trained morphology.

We retain W&B logs, resolved configurations, checkpoint epochs, and local results for compute accounting. Small-scale development and full-scale training costs should be reported separately, including resumed segments and failed launches when discussing practical cost. GPU-hours, rather than epoch counts alone, quantify this budget.

Table 6 reports parameter counts and single-A40 wall-clock cost for the eight corrected 150-clip runs, all completed at the 6,000-epoch budget. Parameter counts are computed analytically from each run’s resolved model configuration (encoder, transformer or flat-MLP, and head widths), not read from a stored checkpoint file, since every linear layer in this codebase is instantiated with `nn.LazyLinear` and has no fixed parameter count outside a materialized model. Parameter count is set by the architecture only, so the refined/unrefined pairs share a count (A/F, B/G, C/E, D/H).

Two consequences run against a naive intuition. First, the temporal MLP has the most total parameters of any architecture, not the fewest: its actor’s first layer reads the full raw concatenation of every observation category (5,248 dimensions) directly, which alone costs more parameters

Table 6: Parameter counts and single-GPU wall-clock cost for the eight corrected 150-clip runs (seed 0, one A40 each, 6,000 epochs). Actor and critic are separate networks with independent parameters.

Case	Arch	Actor params	Critic params	Total params	GPU-h
A / F	temporal MLP	57,349,557	8,524,801	65,874,358	15.8 / 14.4
B / G	attention	45,430,197	5,800,705	51,230,902	23.5 / 21.5
C / E	slot/type	45,434,037	5,804,545	51,238,582	23.7 / 23.9
D / H	slot/type + adaLN	46,256,053	5,804,545	52,060,598	23.9 / 22.4

than the attention models’ three token encoders and transformer combined. The attention models compress that same information to a 256-dimensional token per source before the shared final head, trading parameters for the added transformer computation. Second, despite having more parameters, the temporal MLP is the fastest to train by a wide margin (15.8/14.4 vs. 21.5–23.9 GPU-hours) – parameter count does not predict wall-clock cost here, since the attention models pay for multiple sequential module calls (three encoders, then a transformer stack) rather than one large matrix multiplication chain. Actor-only adaLN-Zero adds a small but non-negligible cost over slot/type (roughly 822,000 actor parameters from the per-layer modulation networks and morphology encoder), consistent with, but not sufficient by itself to explain, the jerk difference in Section 6.3.

The full-scale selected run shares C and E’s architecture exactly (same encoder, transformer, and head widths – `mlp_wide_stage2_discover_attention_slot_type.py` inherits these constants unchanged from the 150-clip experiment file), so its parameter count is also 51,238,582. It ran on six A40 GPUs (`ddp`, 32-bit, 2,048 environments and an 8,192 minibatch per GPU) for roughly 28,170 epochs. The logged rank-0 wall time is about 77 h, of which one interruption near epoch 760 accounts for about 3 h, giving on the order of 450 GPU-hours; this run is not directly comparable to the eight matched single-GPU development runs. An early restart after a memory-related failure resumed from the last checkpoint with the resident pool reduced from 256 to 128 clips per rank and the local cache multiplier set to 1.0; because configs are reloaded from the pickle on resume, the frozen configuration still records the pre-restart pool size of 256, and the earlier failed launch, logged into the same run, is not separately quantifiable pending recovery of the launcher logs. We draw no compute-efficiency claim from this accounting: the temporal MLP’s lower wall-clock cost in Table 6 is not weighed here against its higher position error relative to slot/type in Table 9.

6 Results and Failure Analysis

6.1 Recorded full-scale validation result

Table 7 reports the final scheduled monitoring evaluation recorded for the selected full-scale slot/type run, at epoch 28,159 after approximately 11.07 billion simulated frames. The separately downloaded checkpoint is from epoch 28,170. These values are post-correction, morphology-matched evaluations, not measurements of that later checkpoint or of the independent test split.

The validation score is higher than the resident-pool score, while the reported errors are lower. This is consistent with a curriculum that concentrates residency on difficult clips, so the resident pool is not an independent uniform sample of the training distribution: its 89.84% success rate reflects a difficulty-biased subset of training clips, not the success rate of the entire training corpus, and the gap should not be read as validation outperforming an i.i.d. training sample.

Table 7: Recorded full-scale monitoring snapshot. Validation contains 1,048 held-out clips; each evaluation uses one selected trained body per clip. The active resident pool is difficulty-biased. Values are the stored distributed logging summaries from run `3mvgad6f`, epoch 28,159; no confidence intervals or all-shape coverage are implied.

Metric	Hard resident pool	Validation holdout
Success rate (%) $\uparrow$	89.84	98.47
Mean body-position error (m) $\downarrow$	0.0960	0.0698
Mean global body-rotation error (rad) $\downarrow$	0.2261	0.1760
Mean maximum-joint position error (m) $\downarrow$	0.1848	0.1412
Windowed normalized jerk $\downarrow$	1,442.8	1,151.7
High-jerk windows (%) $\downarrow$	31.05	27.25

Table 8: Offline evaluation of the epoch-28,170 checkpoint on the 1,048-clip validation and 1,048-clip test splits of `hhi_stage2.v1`, one selected trained body per clip (rotating shape panel index 110). Same standalone evaluator for both.

Metric	Validation	Test
Success rate (%) $\uparrow$	98.47	99.14
Failure rate (gt err > 0.5) (%) $\downarrow$	1.53	0.86
Mean body-position error (m) $\downarrow$	0.0699	0.0676
Mean global body-rotation error (rad) $\downarrow$	0.1738	0.1723
Mean maximum-joint position error (m) $\downarrow$	0.1419	0.1393
Normalized jerk $\downarrow$	1,154.3	1,050.9
High-jerk windows (%) $\downarrow$	27.46	25.33
Worst single clip, gt err max (m)	3.79	1.67

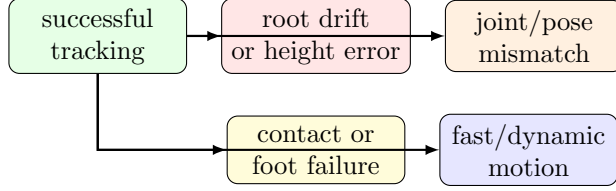
The 98.47% result supports strong tracking of the selected held-out clip/body pairs under the specified threshold, though it speaks only to those pairs: all-body success, unseen-body generalization, and an untouched test-set score are evaluated separately below and in Section 6.5.1. Because a successful rollout can still have visible local errors or jitter, we complement the thresholded success rate with position, rotation, and smoothness measures throughout.

The run record is available at the full-scale W&B run. The current table uses the recorded snapshot, not a newly executed evaluation.

6.2 Standalone validation and held-out test evaluation

The epoch-28,170 checkpoint was re-evaluated offline with a standalone streaming evaluator, once on the validation split and once on the independent test split. Both runs use the same protocol as training-time monitoring – one deterministically selected body shape per clip, an environment with that exact asset, the mean action, and a capped horizon – and write a per-motion record for every clip. The test manifest is loaded only by this standalone evaluator; its SHA-256 and clip count are checked against the split metadata, and its disjointness from train and validation is verified before evaluation. All test clips are the standard 6.6s HumanML3D crop, so none reach the horizon cap. The test split is evaluated once; its number is not iterated against.

Test performance is marginally better than validation on every metric (Table 8), the expected outcome for a held-out reveal: validation was monitored during development while test was not, and the roughly 0.7-point success gap is within the variation expected between two random 5% partitions of the same corpus. As above, the result is scoped to the rotating one-shape-per-clip



Failure analysis uses per-motion records and rendered rollouts; categories are diagnostic hypotheses, not additional success labels.

Figure 5: Failure-analysis taxonomy for the held-out evaluation. The useful unit is the individual clip/body rollout: aggregate success is accompanied by per-motion error traces and visual inspection to distinguish drift, local pose mismatch, contact failures, and difficult dynamic motion.

Table 9: Latest corrected evaluation-fix ablations on the 150-clip development set. Values are the last-eight-evaluation mean $\pm$ standard deviation. Refined-data cases (A–D) are scored against refined references and unrefined-data cases (E–H) against unrefined references; the two halves are not a controlled cross-comparison (Section 6.4). Parameter counts and GPU-hours are from Table 6. Single seed per case.

Case	Architecture / data	Success (%)	gt err	gr err	Jerk	HJ%
A	temporal MLP, refined	97.5 $\pm$ 0.8	0.109	0.207	1,729	33.3
B	attention, refined	97.5 $\pm$ 1.4	0.098	0.228	3,455	40.4
C	slot/type, refined	98.1 $\pm$ 0.6	0.097	0.226	1,757	34.2
D	slot/type + AdaLN, refined	97.1 $\pm$ 1.2	0.090	0.220	3,145	38.9
E	slot/type, unrefined	97.9 $\pm$ 0.6	0.102	0.219	1,422	29.6
F	temporal MLP, unrefined	96.2 $\pm$ 1.6	0.115	0.212	2,927	36.6
G	attention, unrefined	97.8 $\pm$ 1.2	0.098	0.219	3,117	34.9
H	slot/type + AdaLN, unrefined	97.5 $\pm$ 0.9	0.091	0.207	2,956	34.0

protocol, not to all-body coverage or unseen-body generalization. A frozen shape-panel repeat with explicit all-panel coverage for a subset of clips, and public release of the retained per-motion records for both splits, are left for future work.

6.3 Corrected architecture comparison

An earlier evaluator could assign a reference to an environment with the wrong morphology; correcting it changed the meaning of the evaluation, so a rise at the code transition reflects the fix itself rather than instantaneous policy learning. The accompanying sampling correction also changed which training clips entered the resident pool. We distinguish both effects from genuine architectural improvement throughout this section.

The eight corrected runs in Table 5 reassess the 150-clip comparisons using the matching-asset protocol, each for the full 6,000-epoch budget with one training seed. Table 9 reports the mean and standard deviation of each run’s last eight scheduled evaluations (epochs approximately 4,600–6,000); single evaluation points swing by two to four percentage points of success rate and by 100–300 in jerk, so a single final snapshot is not a stable summary.

Success rate does not separate the runs: all eight sit between 96 and 98% with last-8 standard deviations of 0.6–1.6 points, and the intervals overlap. Jerk, and the high-jerk-window percentage

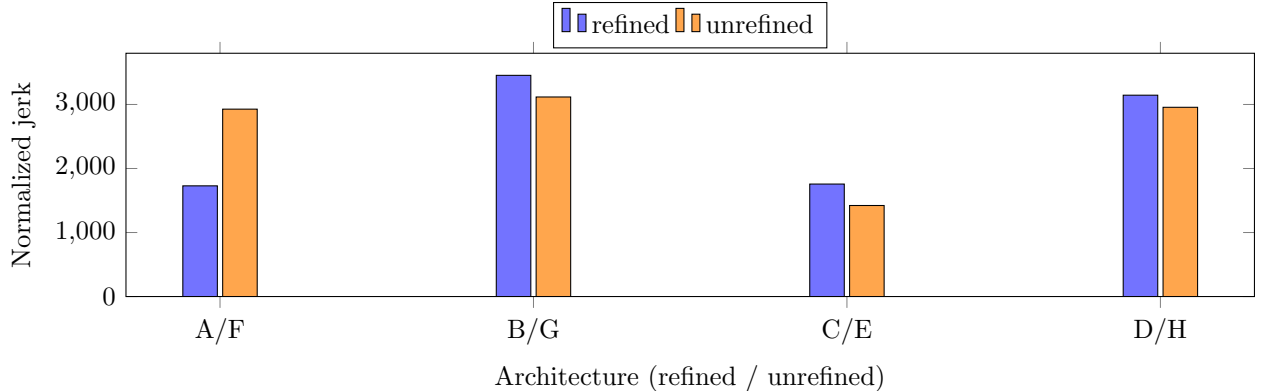


Figure 6: Normalized jerk (last-8-evaluation mean) for the eight 150-clip runs, grouped by architecture. Basic attention (B/G) and slot/type + adaLN-Zero (D/H) roughly double the jerk of the plain temporal MLP (A/F) and of slot/type without adaLN-Zero (C/E), on both reference versions. Success rate is 96–98% for all eight (Table 9) and does not distinguish them.

that tracks it, is the one axis that separates them consistently (Figure 6). On the clean within-data comparison (A–D, identical refined targets), the slot/type run (C) has the highest success rate and low jerk; basic attention (B) and actor-only adaLN-Zero (D) both roughly double the jerk of C, and D’s slightly lower position error is within one standard deviation while its jerk and high-jerk fraction are clearly worse. The temporal MLP (A) matches C on jerk but has clearly higher position error, and is the largest model by parameter count despite training fastest (Table 6). The unrefined group (E–H) reproduces the same ordering: slot/type (E) has the lowest jerk of the four, and adding adaLN-Zero (H) or removing the slot/type embeddings (G) roughly doubles it.

These are single-seed development results, and the pre-correction 150-clip curves are not comparable to them. The load-bearing signal is the jerk difference: basic attention and adaLN-Zero are consistently high, plain MLP and slot/type are consistently low, and this factor-of-two effect holds under both reference versions, so it replicates within the experiment rather than reflecting one lucky run. The success-rate column, by contrast, is non-discriminating. Within those limits the grid supports the selected configuration, slot/type attention without adaLN-Zero, as the best-balanced choice rather than merely the incumbent. Explicit-panel cross-shape numbers for the selected configuration and seed repeats, which would tighten the success-rate column without being expected to move the jerk-based ordering, are left for future work.

6.4 Reference refinement: quality and policy cost

We justify refinement on a deliberately narrow basis: it removes measurable artifacts from the reference motion itself, and training on the refined references does not cost policy quality. We stop short of a causal accuracy benefit from refined training, which the present evidence cannot establish, and evaluate the two parts of the argument separately.

The reference-level part is an offline before/after audit on the packaged motion tensors, with no simulation, over all 19,200 references of the 150-clip development set (150 clips $\times$ 128 body configurations). Table 10 reports two groups of measures: the artifact measures (stance-foot horizontal speed during detected contact, the fraction of frames whose lowest body centre is below the floor, and reference-domain position jerk), and the fidelity measures (per-body geodesic rotation change, root horizontal displacement, and the cross-shape variance of per-shape knee-flexion range, which

Table 10: Reference-refinement audit over all 19,200 development references. Artifact rows: corpus mean before and after, and the fraction of individual motions that improve; the per-motion distributions are in Figure 7. Fidelity rows: corpus mean of the refined-vs-unrefined difference. Jerk is the raw mean third-difference of body positions, not the windowed evaluator score.

	Unref. mean	Refined mean	Motions improved
Stance-foot speed (m/s) ↓	0.055	0.038	96%
Frames below floor (%) ↓	39.6	3.3	45%
Frames within 2 cm of floor (%) ↓	49.8	33.6	—
Position jerk (m/s ³)	45.2	60.7	21%
Per-body rotation change (deg)		0.12	
Root horizontal displacement (m)		0.008	
Cross-shape var., knee-flexion range (ratio r/u)		0.94	

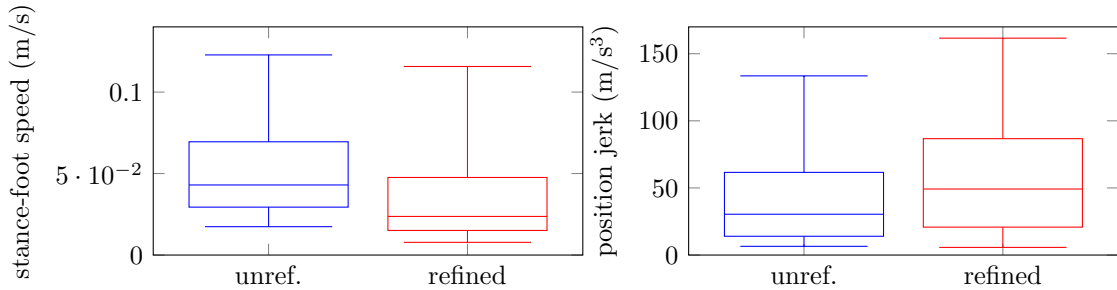


Figure 7: Per-motion distributions (box: quartiles; whiskers: 5th–95th percentile) over the 19,200 development references. Foot-skate drops across almost the whole distribution; position jerk rises across almost the whole distribution.

must be preserved rather than flattened).

Refinement removes most ground penetration and about a third of the stance-foot sliding, while changing joint rotations by a tenth of a degree and the root path by millimetres, and leaving the cross-shape spread of per-shape knee kinematics essentially intact (variance ratio 0.94). The per-motion distributions (Figure 7) show these are not mean-only effects: stance-foot speed falls for 96% of individual motions, across the full range rather than a few outliers, although the worst-sliding 5% of clips are barely improved. Penetration in the raw references is bimodal: a large fraction of clips sit entirely below the floor, a whole-clip vertical mis-registration that the clearance correction removes, while already-grounded clips stay grounded, so only 45% of motions register a change. The cost is a roughly one-third increase in reference-domain position jerk that is broad rather than tail-driven: 79% of motions get jerkier, and the effect is a clip property, not a shape property. Grouped by clip (150), the median jerk moves 30.8 to 50.9 and the range spans 3.9–237 (unrefined) versus 3.4–246 (refined); grouped by body shape (128), the medians move 45.5 to 60.5 within a tight 36–57 versus 51–74 band, so refinement raises jerk almost uniformly across morphologies. It is introduced by the per-window root-translation correction and the per-frame ground-clearance lift. This also explains the earlier observation that the unrefined slot/type run (E) shows lower policy-side jerk than the refined one (C): the refined *reference* itself carries more position jerk, so a policy that tracks it faithfully inherits more, and the E–C gap reflects the target, not a smoother policy from unrefined training.

The policy-cost part reuses the architecture-matched refined and unrefined training pairs from

Table 5 (A/F, B/G, C/E, D/H). Each run is evaluated against its own reference version, so these pairs are not a controlled measurement of a refined-trained policy being more accurate; they show only whether refined training loses anything. Across the four pairs the differences are small and mostly within single-seed noise: the refined run has the lower mean position error in three of four pairs, success rate is within half a percentage point either way in three pairs and clearly higher for the refined run in the temporal-MLP pair, and the jerk difference has inconsistent sign once the reference-jerk offset above is taken into account.

Refined references are therefore used for the full-scale run because they remove most ground penetration and much of the foot sliding, with negligible change to joint angles or root trajectory and no loss of cross-shape variation, and at no measurable policy-accuracy cost. The added reference-domain jerk is a known trade-off. We do not claim a causal full-scale refinement benefit here; establishing one would need a common-reference re-evaluation of the C and E checkpoints, which we leave for future work.

6.5 Cross-shape consistency and qualitative evidence

A useful qualitative comparison places several bodies performing the same clip side by side, with reference markers and a shared time axis; source descriptions can identify the action once mirrored and cropped captions have been checked, and reference-only videos illustrate the generated corpus without standing in for successful policy rollouts. Figure 8 shows a single frame from a recorded simulation video: several of the trained body configurations executing the controller side by side, rendered as the simulator’s collision-geometry capsules rather than SMPL surface mesh. Quantitative cross-shape analysis instead requires the same clip evaluated on an explicit body panel, reporting both aggregate accuracy and variation across shapes – including the number of clips that succeed on every tested body, a statistic the rotating one-shape snapshot used above cannot provide. This explicit-panel evaluation, and the unseen-morphology study below, remain future work.

6.5.1 Unseen-morphology generalization

Training fixes 128 body configurations sampled inside $\beta \in [-3, 3]^{10}$ (Section 3.2). A natural question the trained-body results do not answer is whether the controller has learned morphology-conditioned control, as opposed to memorizing those 128 configurations specifically. Distinguishing the two requires evaluating body shapes drawn from the same coefficient range but excluded from the 128 training combinations. We generated 32 such shapes (16 new beta vectors $\times$ 2 genders, same sampling range, new MJCF assets and HUMOS references) and replayed the checkpoint on all $1,024 \times 32 = 32,768$ (clip, shape) pairs, reusing 1,024 of the corpus’s own clip identities rather than new motions – 932 of them from the training partition, so for those the policy has seen the motion before and only the body is new, isolating the shape axis; the remaining 92 fall in the held-out validation/test partitions, where both motion and shape are unseen simultaneously, and are reported separately below. Rollouts follow the Section 5.4 protocol exactly, and unseen-body success numbers are never mixed into the trained-body results above.

Four of the 32 shapes show severe to near-total tracking failure (Table 11). In each case the identical beta vector applied to the *other* gender’s body template succeeds at 99.8–100%, which points to a gender-specific asset artifact for that specific configuration rather than a genuinely hard region of shape space: a real shape effect would not flip on gender alone while holding the coefficients fixed. We did not confirm the artifact by direct asset inspection, so this remains a morphology-specific anomaly in the sense of Table 12, not a resolved cause, and we report it separately rather

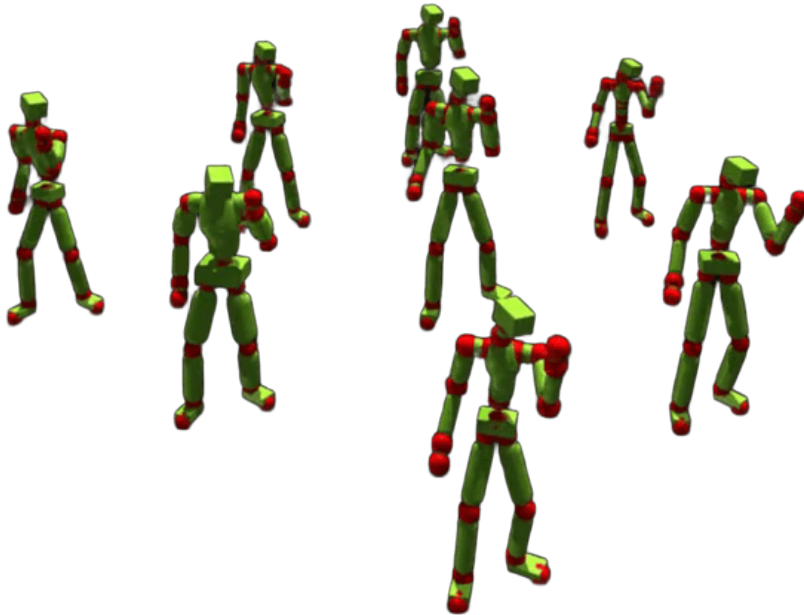


Figure 8: A frame from a recorded simulation video, showing several of the 128 trained body configurations executing the controller simultaneously in parallel environments. Bodies are rendered as the simulator’s collision-geometry capsules (Section 3.2), not the SMPL surface mesh; this illustrates simultaneous multi-body execution and is not a controlled side-by-side comparison on one shared clip.

than let it contaminate the headline generalization claim.

Excluding the four flagged shapes, the remaining 28 succeed at 99.75% on the 932 train-split pairs (mean position error 0.093 m, rotation error 0.200 rad) and 99.61% on the 2,576 validation/test-split pairs where both motion and shape are unseen (mean position error 0.093 m, rotation error 0.204 rad). Both essentially match the trained-body validation and test numbers in Table 8, with no decline across the full 2.58–5.05 distance range evaluated (Figure 9). This is the strong form of the claim in Section 5: for 28 of 32 unseen shapes spanning the full sampled distance range, performance stays flat rather than declining with distance, consistent with morphology-conditioned control rather than memorization of the 128 trained configurations. That claim is scoped to *unseen morphologies within the trained coefficient range*, not to arbitrary body shapes, and the four flagged shapes are a genuine, unresolved anomaly reported alongside it rather than smoothed into the aggregate.

6.6 Evidence-based analysis of residual failures

We treat the observed failure fraction as depending on the evaluated population, shape panel, checkpoint, horizon, and threshold, rather than as a fixed 10% tail or an intrinsic 80% learning ceiling; the old morphology mismatch alone previously produced false failures, so each residual failure is treated as an observation to explain, not as evidence that the motion is physically impossible.

Table 11: The four flagged unseen shapes: the failing gender variant next to its identical-beta counterpart on the other gender (train-split clips, $n = 932$ each). Distance is Euclidean, raw 10-D beta coordinates, to the nearest trained shape of the same gender.

Beta key	Failing variant	Other-gender variant	Distance
a83170f5	male: 58.3%	female: 100.0%	3.24
cf3d5ee7	male: 46.8%	female: 99.8%	3.53
4d1b9df1	female: 35.8%	male: 99.8%	3.53
1d4da893	female: 0.1%	male: 99.8%	3.67

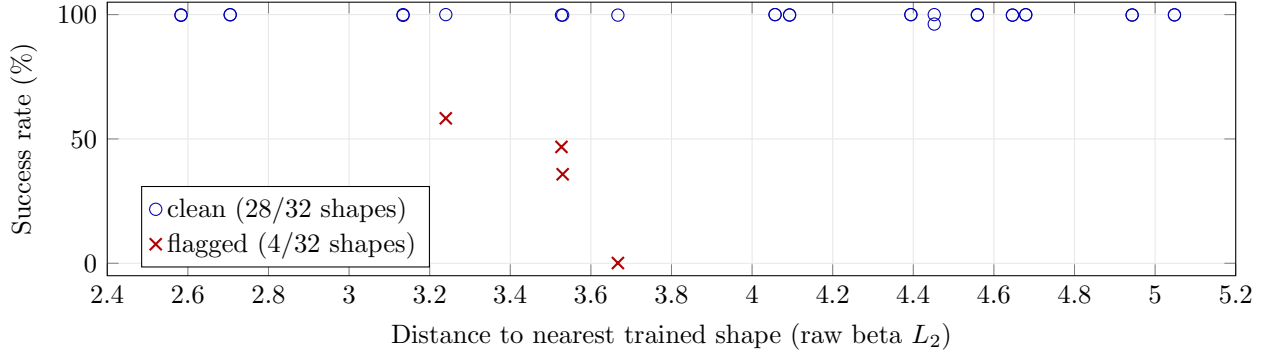


Figure 9: Success rate vs. Euclidean distance to the nearest trained shape (train-split clips, one point per unseen shape, $n = 932$ rollouts per point). The 28 clean shapes show no decline across the full evaluated distance range; the 4 flagged shapes (Table 11) are outliers at every distance, not a tail of a monotonic trend.

The analysis unit is a failed clip/body pair with a reproducible rollout (Figure 5). For each pair, retain the source identity, morphology, first threshold crossing, tracking-error trajectory, and a synchronized reference/controller visualization. Compare it with a successful body variant of the same clip where available. Table 12 defines candidate explanations and the evidence needed to support them.

A complete report should quantify these categories over all evaluated failures, or over a documented stratified sample with appropriate denominators, and include representative successes as controls. Repeated rollouts and targeted interventions can strengthen an explanation; visual inspection alone cannot prove dynamic infeasibility. Capacity limits, reference defects, and body-dependent dynamics may coexist.

6.6.1 Failures on the validation and test splits

Joining the 9 test failures and 16 validation failures (Table 8) to their inherited HumanML3D captions gives a consistent picture across both splits. Every failed clip is a recognizably hard motion; none is an unexplained failure of an ordinary movement. Two motion families account for most of the 25:

- *Transitions to and from the ground* – getting up from lying or prone, belly crawling, push-up postures, sit-to-stand-to-floor: 6 of 9 test failures and 5 of 16 validation failures.
- *Whole-body dynamic rotation* – cartwheels, a running somersault: 2 of 9 test and 4 of 16 validation.

Table 12: Residual-failure analysis framework. Categories are hypotheses until supported by measurements; they can overlap and include an unresolved group.

Candidate explanation	Evidence required
Reference artifact	Sliding, penetration, abrupt rotation/root changes, or inconsistent contacts at the failure time; compare refined and original references.
Missing environmental support	Reference/video and checked caption show an object, seat, wall, partner, or other support absent from the simulated task.
Morphology-specific difficulty	The same clip succeeds on other bodies; compare geometry, clearance, joint limits, and reference differences on the failing body.
Controller or optimization limitation	Repeated failure despite an apparently valid reference; recovery under additional targeted training or an independently tested controller would argue against intrinsic infeasibility.
Metric or protocol effect	Inspect asset identity, horizon, initialization, and threshold crossings; check whether a visually plausible deviation alone changes the binary label.
Unresolved	Retain ambiguous and mixed cases without assigning a physical cause.

The remainder are turning or side-stepping locomotion (5 validation), one clip whose caption describes a hand-held sparring stick that the flat-ground task does not simulate (validation M006386, the single clear missing-support case), one abrupt run-to-stop, and one test clip (M013267) whose peak position error sits exactly at the 0.5 m threshold with a garbled caption, which we treat as a metric-boundary case rather than a tracking failure. Several base motions fail on both their original and mirrored orientation, each on a different body shape (three base motions in test, about six in validation), consistent with a skill-class difficulty that interacts with morphology rather than a single unlucky clip. The ground-transition and acrobatic clusters coincide with the historically hard skill classes seen throughout the development lineage, pointing to a controller or optimization limitation on those classes. The acrobatic failures also carry very high policy-side jerk, and since the corpus-level audit in Section 6.4 shows refinement raises reference jerk on average, a reference artifact on those specific clips remains a live possibility; cross-checking the acrobatic-cluster references against that audit, and confirming the both-orientation failures against successful shapes of the same clip on an explicit panel, are left for future work. We draw no single causal attribution for this mixed cluster.

7 Physical Analysis

This section examines how body morphology changes the physical control demand of a motion, using the simulator’s instrumented quantities and not claims about human biomechanics. The question is not whether body mass or size correlates with joint torque, which is partly expected from mechanics, but whether the effect of morphology on control demand *depends on the motion being performed*.

The procedure holds the motion fixed and varies the body. For the full matched corpus of 150 clips evaluated across all 128 trained shape configurations (19,200 deterministic rollouts total, epoch-28,170 checkpoint, zero failed episodes), we record per-joint torque, joint power, an aggregate control-effort measure, and jerk from the simulator. Clips are grouped by their inherited text descriptions into coarse semantic classes (walking, running, jumping, kicking, dancing, sitting, standing, plus an unclassified *other* bucket) via keyword matching on the concatenated captions, and we compare the cross-morphology variation of each quantity between classes.

The hypothesis under test is that morphological variation produces motion-dependent changes

Table 13: Cross-shape variability of control demand, dynamic (running + jumping + kicking, $n=25$ clips) vs. quasi-static (standing + sitting, $n=37$ clips) motion classes. Each cell is the standard deviation of the pooled, per-clip-demeaned residual (absolute units, clip baseline removed) with a 95% bootstrap CI resampled at the clip level (10,000 resamples). p is Levene’s test (Brown–Forsythe, median-centered) on the same pooled residuals.

Metric	Dynamic std [95% CI]	Quasi-static std [95% CI]	Levene’s p
Normalized torque (frac. effort limit)	1.93 [1.71, 2.21] $\times 10^{-4}$	2.06 [1.86, 2.30] $\times 10^{-4}$	0.10
Mass-normalized power	0.292 [0.220, 0.370]	0.259 [0.210, 0.323]	1.1×10^{-7}
Normalized jerk	350.2 [283.3, 416.9]	403.2 [303.3, 513.5]	0.33

in control demand, with highly dynamic motions (running, jumping, kicking) more sensitive to body shape than slow or quasi-static motions (standing, sitting) – that is, body morphology and motion characteristics interact in setting control demand, rather than morphology acting as a motion-independent scaling.

Instrumentation validation. A first pass compared each class’s per-shape coefficient of variation (CV, std/mean). This produced a highly significant difference ($p < 10^{-15}$ by Levene’s test on every metric) – but in the direction *opposite* the hypothesis, with quasi-static classes appearing more shape-sensitive than dynamic ones. The cause is a normalization artifact rather than a real effect: quasi-static clips have a much smaller mean control demand, so dividing by that small mean inflates relative variance from ordinary simulation noise. We therefore report absolute, unit-preserving residuals instead (each clip’s value minus its own across-shape mean, not divided by it), and we perform inference at the clip level rather than the row level: a clip’s 128 shape replicates are correlated observations of that one clip, not 128 independent draws about its semantic class, so a naive per-row test overstates significance (pseudo-replication).

Result. Table 13 shows no evidence for the hypothesized interaction. For normalized torque and jerk, the dynamic-class and quasi-static-class confidence intervals overlap and Levene’s test is not significant. Mass-normalized power shows a significant Levene’s p , but its clip-level bootstrap CIs overlap and the point estimate is, if anything, marginally *higher* for the dynamic classes – consistent with the row-level pseudo-replication described above rather than a real, class-level effect (the same 128-replicates-per-clip correlation that produced the misleading CV result). Small classes (sitting $n=4$, kicking $n=7$, dancing $n=8$) further limit how finely this test could distinguish individual classes, which is why dynamic and quasi-static are pooled into two groups rather than compared class by class.

Figure 10 shows the same demeaned-residual quantity broken out across all eight semantic classes rather than pooled into the two groups compared above. No class or pair of classes stands out as consistently more shape-sensitive on both metrics; the visibly wider boxes for sitting, kicking, and dancing track their small clip counts, not a distinct physical effect, and are the same limitation noted for Table 13.

We conclude that, for this checkpoint and this metric set, morphology’s effect on control demand does not detectably depend on the motion class being performed: cross-shape variability is statistically indistinguishable between highly dynamic and quasi-static motions once the near-zero-baseline normalization artifact is corrected and inference is done at the clip level. This is a properly negative result rather than an absence of analysis: the instrumentation is validated (magnitudes match the tracking-jerk figures already reported in this paper, and the effort-limit normalization

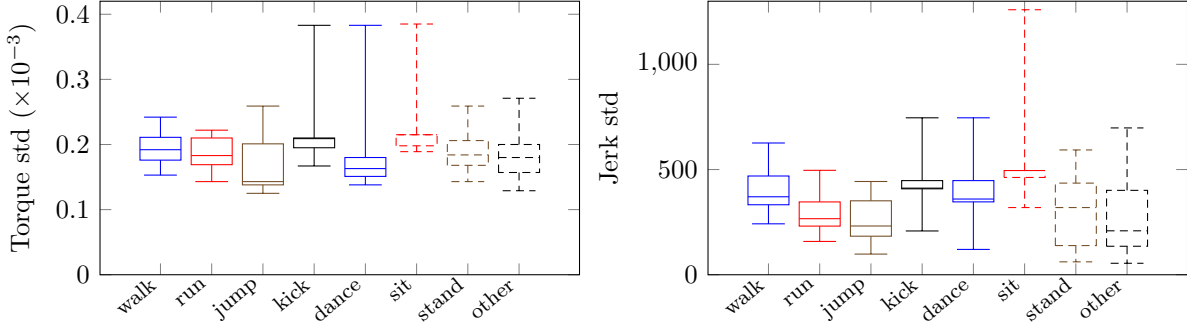


Figure 10: Per-clip cross-shape standard deviation (absolute units; box: quartiles, whiskers: 5th–95th percentile), pooled by semantic class rather than the two pooled groups in Table 13 – the per-class distributions the paper’s own framing calls for. Classes overlap heavily on both metrics with no separation between the dynamic classes (run, jump, kick) and the quasi-static ones (sit, stand); the smallest classes (sit $n=4$, kick $n=7$, dance $n=8$) are visibly wider and noisier, consistent with their limited sample size rather than a distinct effect.

reuses the same mass-scaling formula validated elsewhere in this work), and the same pipeline that would have reported a false positive under a naive CV comparison instead surfaces its own artifact under a correct, clip-level test. No mechanical or biomechanical claim is made beyond what these instrumented rollouts show for this policy.

8 Discussion and Limitations

The current evidence supports the feasibility of combining shape-specific references, matching simulation assets, and one shared controller at the scale of the constructed corpus. The corrected 150-clip grid selects slot/type attention without adaLN-Zero, on the basis of a large, replicated jerk difference rather than the non-discriminating success-rate column, and this ordering is single-seed. The refinement audit shows the processed references have substantially less penetration and foot sliding at negligible fidelity cost, but we stop short of a causal learning benefit from refined training: the refined and unrefined runs track different targets and were not cross-evaluated on a common reference.

The physical analysis in Section 7 adds a cautionary result about the choice of spread statistic itself. A coefficient-of-variation comparison, an otherwise natural way to normalize cross-shape variability by each clip’s own baseline demand, produced a highly significant difference in exactly the direction opposite the hypothesis, because it divides by a small number for quasi-static clips. Only a clip-level test on absolute, unit-preserving residuals revealed this as an artifact rather than a real effect. We surface this as a methodological point for similar cross-body comparisons, independent of whether the underlying interaction hypothesis holds for other checkpoints or corpora.

The dataset is derived from existing motion capture, captions, body models, and HUMOS generation. The 2,681,728 variants represent combinations of 20,951 clip identities and 128 configurations, not equally many independent human examples. Mirrored clips are also not independent captures. Caption alignment, left/right semantics, body-specific wording, and redistribution permissions require further auditing before a benchmark release or language-grounded performance claim.

Refinement changes kinematics rather than solving the complete equations of motion. Ground

clearance of collision proxies does not guarantee balanced support, plausible contact forces, or realistic joint torques. Similarly, estimated asset masses and mass-scaled actuators are modeling choices rather than a calibrated description of human physiology. The flat-ground task excludes much environmental interaction, and text-based filtering cannot ensure that every remaining clip satisfies that scope.

Generalization is limited by the split and evaluation design. Original and mirrored variants stay together, but the current split is not demonstrated to be subject- or source-recording-disjoint. Validation reuses the trained body set and has been monitored during development. Its rotating panel covers only one shape per clip at a time, and long clips have a capped evaluation horizon. The unseen-morphology test in Section 6.5.1 supports morphology conditioning over memorization for 28 of 32 held-out shapes, but the remaining 4 are a genuine, unresolved anomaly, and an explicit all-panel evaluation of the 128 trained shapes remains future work. Any morphology-generalization statement is confined to body shapes inside the trained coefficient range rather than arbitrary bodies.

The success threshold is permissive relative to fine-grained pose accuracy. A high success rate should be read together with continuous errors, smoothness, and qualitative rollouts. The logged normalized-jerk measure also has implementation-specific windows and regularizers; it is not a standalone certificate of physical plausibility. Raw policy-action increments must not be interpreted as joint-angle or torque measurements.

Cost constraints motivated small-scale development and streamed full-scale training, and introduced memory-related resumes and configuration changes on the full-scale run. The 150-clip grid is single-seed: the architecture conclusion is drawn from the jerk difference, which is a large effect that recurs across both reference versions, and not from the success-rate differences, which are within run-to-run noise and are reported as non-discriminating. Seed repeats would sharpen the latter but are not expected to change the former. These costs and choices should be reported transparently; they do not by themselves establish better compute efficiency than another method.

We explored neutral-body pretraining followed by multi-shape transfer and mixed canonical/HUMOS source-weighted rewards during development, but did not retain either as the main approach. Their historical measurements are not used as controlled negative results, particularly where the earlier evaluation protocol differed. The present method is direct multi-shape training. Future work may revisit these alternatives after the current comparisons and dataset audits are complete.

9 Conclusion

We have described a text-associated multi-morphology motion corpus and a physics-based imitation system that preserves the correspondence between source clips, generated references, and simulated bodies. The construction combines AMASS/HumanML3D provenance, HUMOS generation, SMPLSim assets, conservative reference refinement, and deterministic clip-grouped splits. The controller uses temporal slot/type attention with morphology information in both actor and critic and is trained directly across body configurations.

Offline evaluation of the selected checkpoint gives 98.47% success on validation and 99.14% on the untouched test split for the one-shape-per-clip trained-body protocol, and the eight-run development grid selects slot/type attention without adaLN-Zero. Residual failures on both splits are dominated by ground-contact transitions and whole-body acrobatics. Replaying this checkpoint across all 19,200 (clip, shape) pairs of the development corpus finds no statistically detectable evidence that dynamic motions are more shape-sensitive to body morphology than quasi-static ones,

once a near-zero-baseline normalization artifact in the naive comparison is corrected – a properly negative result rather than an unresolved question. Replaying the checkpoint on 32 body shapes excluded from training entirely, sampled in the same coefficient range, finds that 28 of the 32 track successfully at the trained-body level with no decline across the evaluated distance from training – support for morphology-conditioned control over memorization of the 128 trained configurations – while the remaining 4 fail severely in a pattern that flips with gender at fixed shape coefficients, consistent with a gender-specific asset artifact rather than a genuine generalization limit; we report this anomaly rather than average it away. These results do not establish all-shape coverage: an explicit all-panel evaluation of the 128 trained shapes remains future work.

A Reproducibility Details

A.1 Source preparation and refinement settings

The AMASS source collections used by the preparation pipeline are ACCAD, BioMotionLab_NTroje, BMLhandball, BMLmovi, CMU, DFaust.67, EKUT, Eyes_Japan_Dataset, HumanEva, KIT, MPI_HDM05, MPILimits, MPI_mosh, SFU, SSM_synced, TCD_handMocap, TotalCapture, and Transitions_mocap. Source preparation operates at 20 Hz; the final motion libraries use 30 Hz. Joint ordering and quaternion conventions must follow the matching asset and MotionLib conversion rather than be inferred from source array positions.

Table 14: Default settings of the current reference-refinement procedure. Frame counts refer to the 30 Hz MotionLib representation. These are kinematic-processing settings, not physical constraints learned by the policy.

Operation	Setting
Contact candidates	Left/right ankle and toe bodies
Contact detection thresholds	Speed 0.15 m/s; height 0.10 m
Contact cleanup	Close gaps up to 2 frames; remove runs shorter than 3 frames
Rotation filter	Savitzky–Golay, 9-frame window, polynomial order 3
Rotation blend / bound	Strength 0.5; maximum geodesic correction 8°
Contact-transition guard	Separate segments; 2-frame transition taper
Root XY correction	Median stance anchors; least-squares common translation
Root correction regularization	21-frame smoothing window; 0.10 m displacement cap
Ground clearance	0.005 m collision-geometry margin; 5-frame lift smoothing
Final reconstruction	Per-asset FK and per-motion velocity recomputation

A.2 PD gains and limits

Table 15 gives the base values configured for the SMPL morphology robot. For joint d belonging to rigid body j , the simulator applies the multiplier

$$\rho_{b,j} = \frac{m_{b,j}}{m_{\text{ref},j}} \quad (12)$$

to stiffness, damping, and effort limit. The velocity limit is not mass-scaled. All relevant asset masses, joint ranges, collision settings, and the reference asset must accompany a reproducible physical interpretation.

Table 15: Base PD and actuation values before per-body mass scaling. Stiffness and damping use the simulator’s rotational PD units (Nm/rad and Nms/rad); effort limits are in Nm and velocity limits in rad/s.

Joint family	Stiffness	Damping	Effort limit	Velocity limit
Hip, knee, ankle	800	80	500	100
Toe	500	50	500	100
Torso, spine, chest	1,000	100	500	100
Neck, head, thorax, shoulder, elbow	500	50	500	100
Wrist, hand	300	30	500	100

A.3 Windowed jerk convention

The logged smoothness score is computed from simulated global body positions using consecutive finite differences with control interval $\Delta t = 1/30$ s. A nominal 0.4 s window contains $W = \max(4, \lfloor 0.4/\Delta t \rfloor)$ frames, giving $W = 12$ at the configured rate and duration $T_w = (W - 1)\Delta t$. For body j in window w , let $L_{w,j}$ be the sum of speed times Δt and let $\ddot{p}_{w,j}$ denote the third finite difference. The implementation uses

$$J_{w,j} = \frac{T_w^5 \sum_{t \in w} \|\ddot{p}_{t,j}\|_2^2 \Delta t}{\max(L_{w,j}, \epsilon)^2 + \epsilon}, \quad \epsilon = 0.1, \quad (13)$$

where the sum includes the valid third-difference samples in the window. Window scores are averaged per body and then across bodies and motions. High-jerk percentage counts windows for which $\max_j J_{w,j} > 6,500$.

Equation (13) documents the existing numerical convention, including its regularizers; it is not presented as a universally dimensionless or sampling-rate-invariant biomechanical measure. Comparisons must keep the sample rate, window, units, regularizers, and aggregation fixed. Reference refinement reports also include raw jerk in m/s^3 ; those values are a different metric and cannot be compared numerically to this logged score.

A.4 Split and experiment provenance

The split metadata is stored in `data/splits/hhi_stage2_v1/split_metadata.json`, with schema version 1, split seed 42, and split identifier

`4c1c9d592e3dbeef15abe18cb50a3141820735849414ecc2d8959e9f38502263`

It records the source-manifest hash, training/validation/test manifest and ID-list hashes, difficulty information, and the forced training reservation. The generator uses 100 difficulty strata; the 150 requested development identities expand to 286 retained identities when existing mirrored partners are included. Checkpoint/run split provenance refers to these artifacts, not just a verbal 90/5/5 ratio.

The documented HUMOS configuration selects `logs/humos/q6zbv2tu/checkpoints/latest-epoch=1599.ckpt`. The immutable HUMOS checkpoint checksum and the SMPLSim / asset-generation revision hashes are to be archived with the dataset release; a checkpoint path identifies the local workflow but is not a substitute for a checksum.

The principal implementation entry points are:

- Reference refinement: `tools/refine_humos_motion.py`; streamed per-clip processing: `tools/refine_humos_r2_per_clip.py`.

- Selected full-scale experiment (`examples/experiments/mimic/`): `mlp_wide_stage2_discover_attention_slot_type.py`, inheriting the common attention experiment and enabling slot/type embeddings.
- Corrected development comparison: `tools/train_150_evalfix_ablation.sh` (cases A–E) and `tools/train_150_evalfix_unrefined.sh` (cases F–H), seed 0.
- Motion residency and evaluation: `protomotions/components/global_clip_pool.py` and `protomotions/agents/evaluators/mimic_evaluator.py`.
- Standalone split evaluation: `evaluate_validation_split.py` and `evaluate_test_split.py` under `protomotions/`. Each output records the checkpoint, split, reference version, and shape panel; the test evaluator additionally checks the test-manifest hash, count, and disjointness before running.
- Physical analysis (Section 7): semantic classification in `tools/classify_clip_semantics.py`, the instrumented replay in `tools/analyze_morphology_motion_interaction.py`, and the clip-level significance test in `tools/test_morphology_motion_variance_significance.py`.
- Unseen-morphology generalization (Section 6.5.1): held-out shape assets in `protomotions/data/assets/mjcf/smpl_mor_interp/` (generation pipeline in `note/README.heldout-pipeline.md`), the replay in `tools/analyze_unseen_morphology_generalization.py`, and the distance/split join in `tools/aggregate_unseen_morphology_generalization.py`.

For each final experiment, retain the resolved configuration, code revision, model/checkpoint checksum, body coefficient and asset manifests, split hashes, all resume segments, frame count, GPU model/count, and elapsed GPU-hours. For each evaluation, retain per-pair clip and asset IDs, panel selection, success/failure, valid horizon, and component errors in a local analysis file. The main full-scale W&B identifier is `3mvgad6f`; the corrected small-scale group is `hhi_150_evalfix_comparison`. The evaluation/sampling correction is recorded in commit `3685247`.

Two items remain before a finalized benchmark report: archiving the immutable HUMOS/SMPLSim checksums alongside the dataset release, and releasing the retained per-motion evaluation records for the validation and test splits.

References

- [1] Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael J. Black. AMASS: Archive of Motion Capture as Surface Shapes. In *ICCV*, 2019. <https://arxiv.org/abs/1904.03278>.
- [2] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating Diverse and Natural 3D Human Motions From Text. In *CVPR*, 2022. CVF Open Access paper.
- [3] Shashank Tripathi, Omid Taheri, Christoph Lassner, Michael J. Black, Daniel Holden, and Carsten Stoll. HUMOS: Human Motion Model Conditioned on Body Shape. In *ECCV*, 2024. <https://arxiv.org/abs/2409.03944>.
- [4] Xue Bin Peng, Pieter Abbeel, Sergey Levine, and Michiel van de Panne. DeepMimic: Example-Guided Deep Reinforcement Learning of Physics-Based Character Skills. *ACM Transactions on Graphics*, 37(4), 2018. <https://xbpeng.github.io/projects/DeepMimic/index.html>.

- [5] Zhengyi Luo, Jinkun Cao, Alexander W. Winkler, Kris Kitani, and Weipeng Xu. Perpetual Humanoid Control for Real-time Simulated Avatars. In *ICCV*, 2023. CVF Open Access paper.
- [6] The ProtoMotions contributors. ProtoMotions. Open-source software for physics-based character animation and humanoid control. <https://github.com/NVlabs/ProtoMotions>.
- [7] Viktor Makoviychuk, Lukasz Wawrzyniak, Yunrong Guo, Michelle Lu, Kier Storey, Miles Macklin, David Hoeller, Nikita Rudin, Arthur Allshire, Ankur Handa, and Gavriel State. Isaac Gym: High Performance GPU-Based Physics Simulation For Robot Learning. Technical report, 2021. <https://arxiv.org/abs/2108.10470>.
- [8] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal Policy Optimization Algorithms. Technical report, 2017. <https://arxiv.org/abs/1707.06347>.
- [9] John Schulman, Philipp Moritz, Sergey Levine, Michael I. Jordan, and Pieter Abbeel. High-Dimensional Continuous Control Using Generalized Advantage Estimation. In *ICLR*, 2016. <https://arxiv.org/abs/1506.02438>.
- [10] Jungdam Won and Jehee Lee. Learning Body Shape Variation in Physics-based Characters. *ACM Transactions on Graphics*, 38(6), Article 207, 2019. <https://doi.org/10.1145/3355089.3356499>.
- [11] Matthew Loper, Naureen Mahmood, Javier Romero, Gerard Pons-Moll, and Michael J. Black. SMPL: A Skinned Multi-Person Linear Model. *ACM Transactions on Graphics*, 34(6), 2015. <https://smpl.is.tue.mpg.de/>.
- [12] Zhengyi Luo and contributors. SMPLSim. Open-source SMPL-family body-model simulation software. <https://github.com/ZhengyiLuo/SMPLSim>.
- [13] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In *NeurIPS*, 2017. <https://arxiv.org/abs/1706.03762>.
- [14] William Peebles and Saining Xie. Scalable Diffusion Models with Transformers. In *ICCV*, 2023. <https://arxiv.org/abs/2212.09748>.
- [15] Zhengyi Luo, Jinkun Cao, Josh Merel, Alexander Winkler, Jing Huang, Kris Kitani, and Weipeng Xu. Universal Humanoid Motion Representations for Physics-Based Control. In *ICLR*, 2024. <https://arxiv.org/abs/2310.04582>.
- [16] Tairan He, Zhengyi Luo, Xialin He, Wenli Xiao, Chong Zhang, Weinan Zhang, Kris Kitani, Changliu Liu, and Guanya Shi. OmniH2O: Universal and Dexterous Human-to-Humanoid Whole-Body Teleoperation and Learning. In *CoRL*, 2024. <https://arxiv.org/abs/2406.08858>.
- [17] Mazeyu Ji, Xuanbin Peng, Fangchen Liu, Jialong Li, Ge Yang, Xuxin Cheng, and Xiaolong Wang. ExBody2: Advanced Expressive Humanoid Whole-Body Control. Technical report, 2024. <https://arxiv.org/abs/2412.13196>.
- [18] Zixuan Chen, Mazeyu Ji, Xuxin Cheng, Xuanbin Peng, Xue Bin Peng, and Xiaolong Wang. GMT: General Motion Tracking for Humanoid Whole-Body Control. Technical report, 2025. <https://arxiv.org/abs/2506.14770>.

- [19] Wenlong Huang, Igor Mordatch, and Deepak Pathak. One Policy to Control Them All: Shared Modular Policies for Agent-Agnostic Control. In *ICML*, 2020. <https://proceedings.mlr.press/v119/huang20d.html>.
- [20] Xue Bin Peng, Ze Ma, Pieter Abbeel, Sergey Levine, and Angjoo Kanazawa. AMP: Adversarial Motion Priors for Stylized Physics-Based Character Control. *ACM Transactions on Graphics*, 40(4), 2021. <https://arxiv.org/abs/2104.02180>.
- [21] Xue Bin Peng, Yunrong Guo, Lina Halper, Sergey Levine, and Sanja Fidler. ASE: Large-Scale Reusable Adversarial Skill Embeddings for Physically Simulated Characters. *ACM Transactions on Graphics*, 41(4), 2022. <https://arxiv.org/abs/2205.01906>.
- [22] Chen Tessler, Yoni Kasten, Yunrong Guo, Shie Mannor, Gal Chechik, and Xue Bin Peng. CALM: Conditional Adversarial Latent Models for Directable Virtual Characters. In *SIGGRAPH*, 2023. <https://arxiv.org/abs/2305.02195>.
- [23] Chen Tessler, Yunrong Guo, Ofir Nabati, Gal Chechik, and Xue Bin Peng. MaskedMimic: Unified Physics-Based Character Control Through Masked Motion Inpainting. *ACM Transactions on Graphics*, 43(6), 2024. <https://arxiv.org/abs/2409.14393>.
- [24] Pei Xu, Kaixiang Xie, Sheldon Andrews, Paul G. Kry, Michael Neff, Morgan McGuire, Ioannis Karamouzas, and Victor Zordan. AdaptNet: Policy Adaptation for Physics-Based Character Control. *ACM Transactions on Graphics*, 42(6), 2023. <https://arxiv.org/abs/2310.00239>.
- [25] Zheng Xiong, Jacob Beck, and Shimon Whiteson. Universal Morphology Control via Contextual Modulation. In *ICML*, 2023. <https://arxiv.org/abs/2302.11070>.
- [26] Zheng Xiong, Risto Vuorio, Jacob Beck, Matthieu Zimmer, Kun Shao, and Shimon Whiteson. Distilling Morphology-Conditioned Hypernetworks for Efficient Universal Morphology Control. In *ICML*, 2024. <https://arxiv.org/abs/2402.06570>.
- [27] Laurens Engwegen, Max Weltevrede, Caroline Horsch, Daan Brinks, and Wendelin Böhmer. Shared Modular Recurrence in Contextual MDPs for Universal Morphology Control. Technical report, 2025. <https://arxiv.org/abs/2506.08630>.
- [28] He Zhang, Sebastian Starke, Taku Komura, and Jun Saito. Mode-Adaptive Neural Networks for Quadruped Motion Control. *ACM Transactions on Graphics*, 37(4), 2018. <https://doi.org/10.1145/3197517.3201366>.
- [29] Sebastian Starke, Ian Mason, and Taku Komura. DeepPhase: Periodic Autoencoders for Learning Motion Phase Manifolds. *ACM Transactions on Graphics*, 41(4), 2022. <https://doi.org/10.1145/3528223.3530178>.
- [30] Matthias Plappert, Christian Mandery, and Tamim Asfour. The KIT Motion-Language Dataset. *Big Data*, 4(4), 2016. <https://doi.org/10.1089/big.2016.0028>.
- [31] Abhinanda R. Punnakal, Arjun Chandrasekaran, Nikos Athanasiou, Alejandra Quiros-Ramirez, and Michael J. Black. BABEL: Bodies, Action and Behavior with English Labels. In *CVPR*, 2021. <https://arxiv.org/abs/2106.09696>.
- [32] Jing Lin, Ailing Zeng, Shunlin Lu, Yuanhao Cai, Ruimao Zhang, Haoqian Wang, and Lei Zhang. Motion-X: A Large-Scale 3D Expressive Whole-Body Human Motion Dataset. In *NeurIPS*, 2023. <https://arxiv.org/abs/2307.00818>.

- [33] Mathis Petrovich, Michael J. Black, and Gül Varol. TEMOS: Generating Diverse Human Motions from Textual Descriptions. In *ECCV*, 2022. <https://arxiv.org/abs/2204.14109>.
- [34] Chuan Guo, Xinxin Zuo, Sen Wang, and Li Cheng. TM2T: Stochastic and Tokenized Modeling for the Reciprocal Generation of 3D Human Motions and Texts. In *ECCV*, 2022. <https://arxiv.org/abs/2207.01696>.
- [35] Guy Tevet, Sigal Raab, Brian Gordon, Yoni Shafir, Daniel Cohen-Or, and Amit H. Bermano. Human Motion Diffusion Model. In *ICLR*, 2023. <https://arxiv.org/abs/2209.14916>.
- [36] Mingyuan Zhang, Zhongang Cai, Liang Pan, Fangzhou Hong, Xinying Guo, Lei Yang, and Ziwei Liu. MotionDiffuse: Text-Driven Human Motion Generation With Diffusion Model. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(6), 2024. <https://arxiv.org/abs/2208.15001>.